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Final Work Report

PROJECT TITLE- COLLISION OF MOBILE EQUIPMENT OR HIT BY MACHINE IN UG MINES

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2. Executive Summary

Underground mining is an industry that poses a dangerous workplace with operators and heavy machinery working in enclosed spaces with reduced visibility, dust, darkness, and obstructed views. The large vehicles used in mining operations have considerable blind spots, which can complicate efforts to see any other person or obstacle nearby, leading to accidents. This makes the use of proximity detection and early warning systems vital.

The current technologies include proximity detection using radar technology, laser distance sensors, and wearable sensors; however, false alarms, limitations, lack of effective communication, and costly industrial safety systems make it necessary to develop cheaper solutions.

This project seeks to develop an inexpensive two path collision warning system.

First, the detection circuit is attached to the machine. The hardware consists of a DFRobot C4001 mm-wave radar for detection of human proximity and a VL53L0X ToF sensor for measuring the distance to nearby objects. In case of a detected obstacle located within the predetermined range of 1.5 meters, the system triggers a buzzer and an LED while displaying the results on the OLED display. Photos are taken by an ESP32-CAM module and stored in JPEG format on an SD card. A heartbeat algorithm ensures that the camera node and its storage works correctly.

The second detection path consists of a wearable worker unit with an ESP32-C3 Super Mini and a Reyax RYUW122 ultra-wideband UWB module. An appropriate UWB tag is attached to the machine for distance measurement between the worker and the vehicle.

The two links are intended to use ESP-NOW to exchange information, allowing direct communication between ESP32 devices without any need for a router or internet connection. This approach is expected to increase the likelihood of warning availability in case of failures in either of the detection links or the communication itself.

The machine-mounted sensor and camera system has been tested in the lab, and UWB ranging has been enabled and stabilized with the help of filtering and validation algorithms. Two-way communication firmware is now available for an integrated field test. The wearable unit at the moment generates audible and visual warnings; its vibration actuator is still dysfunctional despite the replacement.

The prototype is a working example only, not a certified mining safety system. Its detection capabilities, communication performance, and warning ranges must be tested integrated. More specifically, the 1.5 meters warning range that was used in bench testing is not sufficient for mining vehicles without accounting for their speed and stopping distance.

3. Introduction

Mining is one of those industries where huge machines and pedestrians exist side by side. A haul truck or dumper may measure up to several metres in width and height, and its operator cannot observe everything going on around him/her. Turning around is even more dangerous since the immediate rear of the machine is the most obscured part of the whole vehicle. Add to this dust, wetness, night time operations and darkness in the underground mine, and a pedestrian may easily become unnoticeable to the driver.

There are facts to prove that this problem is real and present. The USA Mine Safety and Health Administration (MSHA) found out that proximity detection systems would have been able to avoid 42 fatalities caused by accidents with coal haulage machines and scoops from 1984 till 2014. It is known that 42 people died and 179 more were injured during this period of time due to the absence of proximity detection systems [1][2]. Indian Directorate General of Mines Safety (DGMS) also constantly sends safety alerts about such incidents. One of the alerts released in 2017 talks about a supervisor killed by a 100-tonne dumper going after the same shovel he was working with [7].

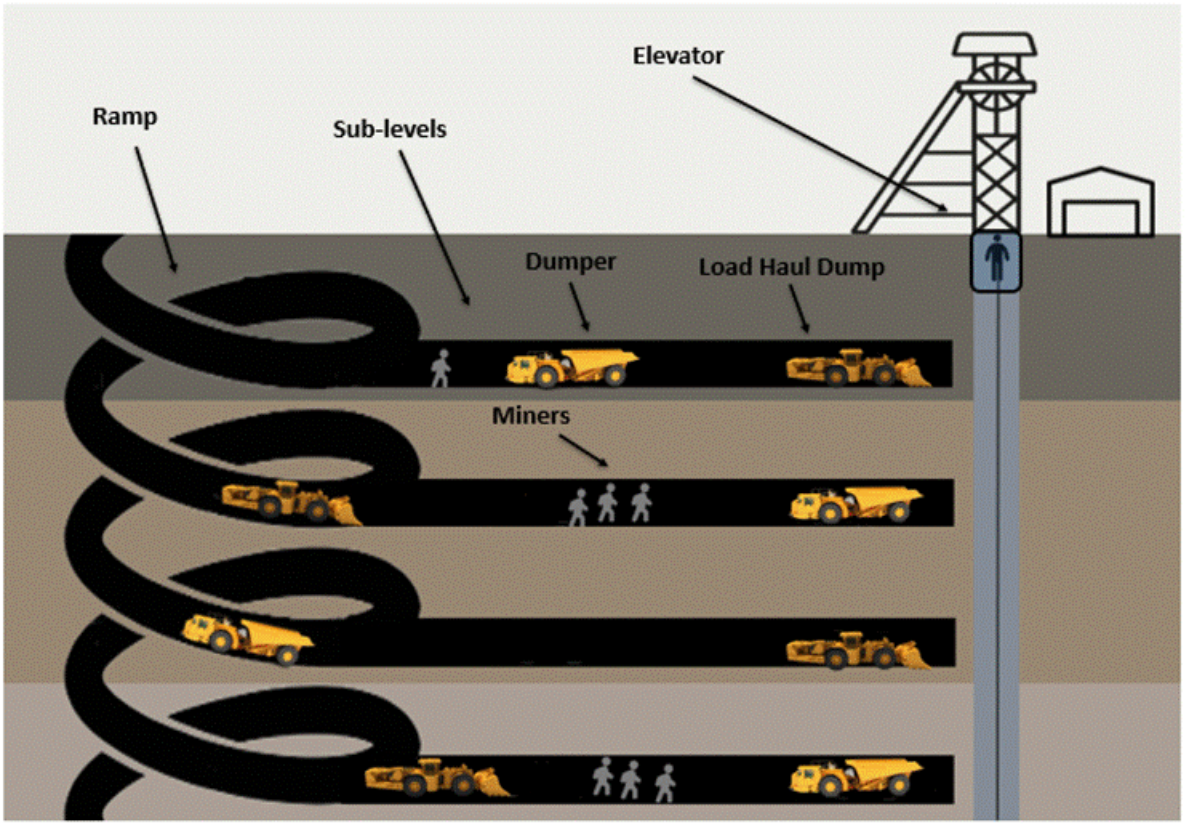


Fig:1 Underground mine description.

Underground mining involves several types of machinery for loading, transportation, drilling, excavation, ground support, and personnel movement. Equipment such as Load-Haul-Dump (LHD) machines, underground haul trucks, drilling jumbos, continuous miners, roadheaders, roof bolters, personnel carriers, and utility vehicles are commonly involved in underground operations. These machines operate in restricted spaces where limited visibility and narrow tunnels can increase the possibility of accidents.

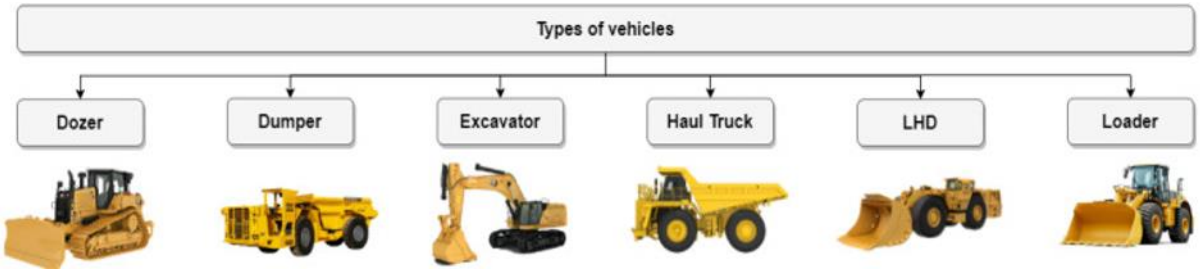


Fig:2 Shows different types of mining machinery used either in surface mining or in underground mining.

One of the major safety challenges in underground mines is the possibility of collisions involving mobile mining equipment. Common accidents include vehicle–worker collisions, vehicle–vehicle collisions, vehicle–wall or rib collisions, vehicle–obstacle collisions, rear-end collisions, turning or intersection collisions, reversing collisions, vehicle–infrastructure collisions, and run-over or crushing accidents. These accidents may occur because of blind spots, poor lighting, narrow tunnels, limited visibility, obstacles, or operator error.

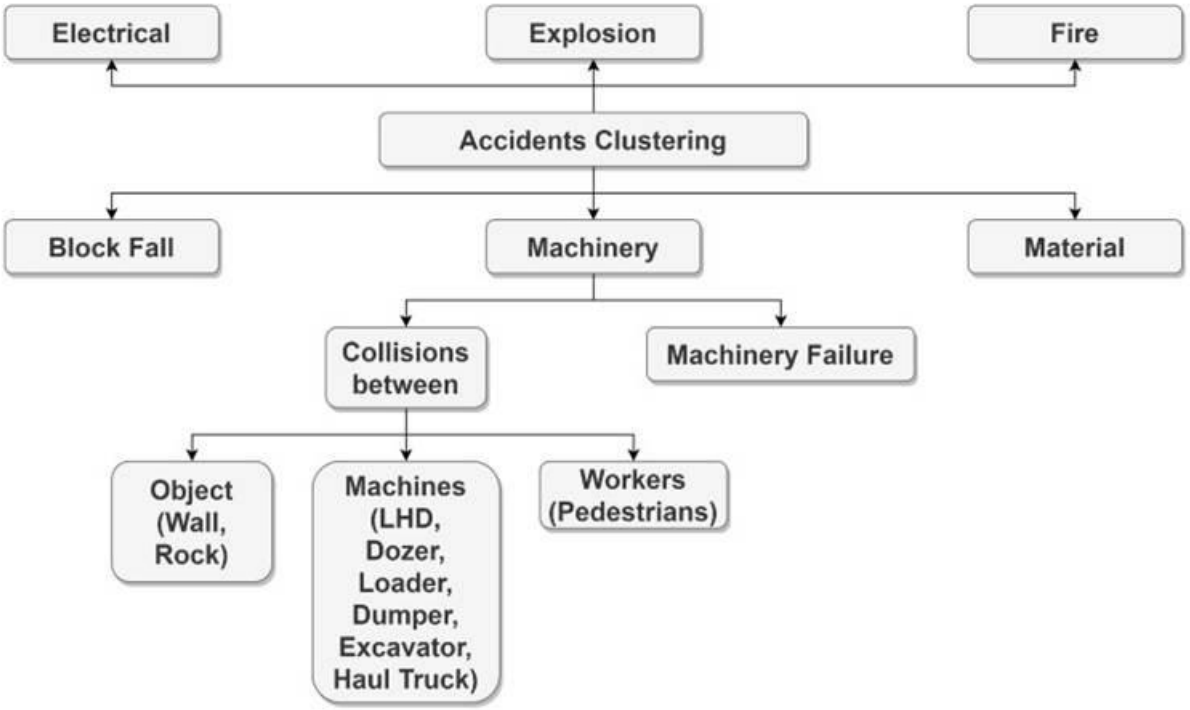


Fig:3 Accident clustering.

A particularly important problem is the presence of blind spots around heavy mining vehicles. Visibility studies involving equipment such as dump trucks, loaders, earth compactors, and excavators show that visibility can vary significantly around different sectors of the machine. The rear and side areas of LHD-type equipment can contain particularly dangerous blind zones. A worker may remain within several metres of a machine while still being invisible to the operator.

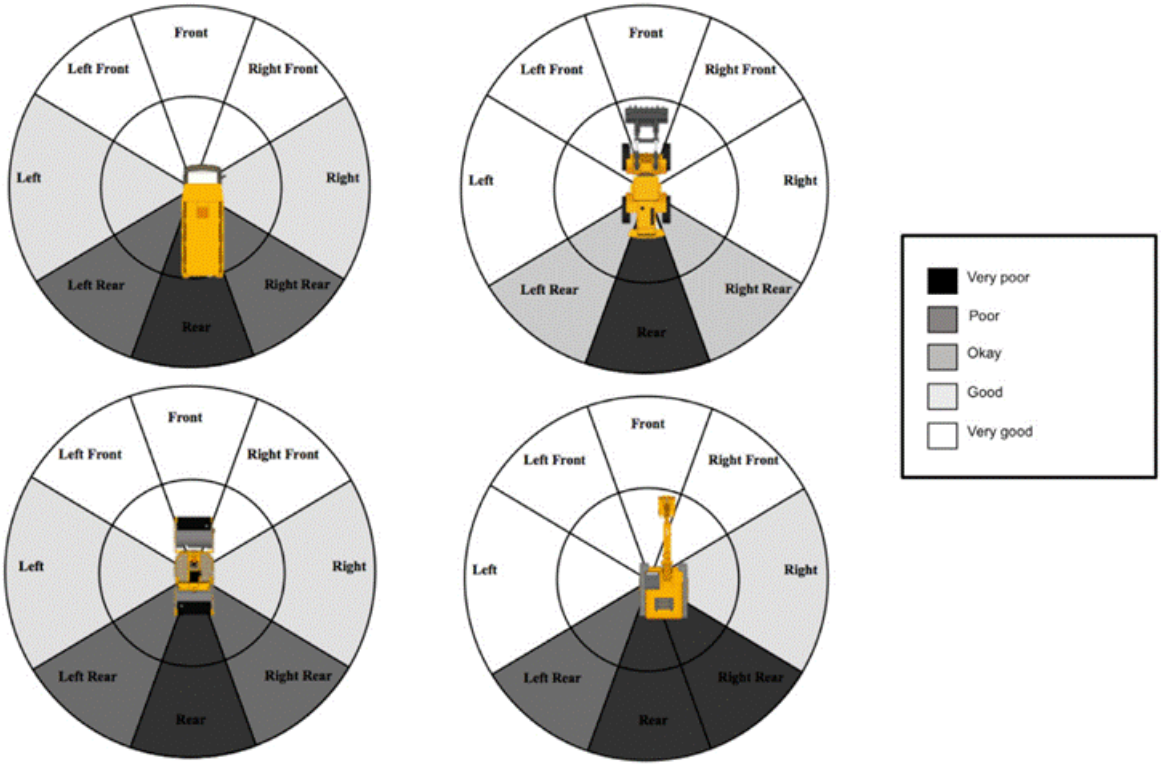


Fig:4 Visibility ratings for each piece of equipment (dump truck, loader, earth compactor, excavator).

Several technologies have been developed to address these safety problems. Proximity Detection Systems use technologies such as RFID, UWB, and magnetic fields to detect workers near machines. Collision Avoidance Systems can combine UWB, radar, and decision-making logic to provide warnings or stop machinery. Radar-based systems can operate in dust, smoke, and low-light environments, while LiDAR can provide accurate distance and spatial information. Thermal cameras can detect human presence in darkness, whereas camera-based computer vision can provide object-classification capabilities. However, these technologies have individual limitations, including false alarms, dust sensitivity, high cost, tunnel reflections, limited object classification, and dependence on environmental conditions.

Therefore, a reliable underground mine safety system requires the integration of multiple sensing technologies rather than depending on a single sensor. The proposed project addresses this requirement through a multi-sensor collision-prevention system that combines **Radar/LiDAR, VL53L0X time-of-flight sensor (I2C), and a UWB-enabled wearable safety tag**. Radar/LiDAR is used to detect humans and estimate their distance, while the VL53L0X time-of-flight sensor (I2C) measures the distance to any obstacle. The UWB wearable tag provides accurate distance or location information and enables bidirectional protection between the worker and the mobile equipment.

3.2 Problem statement

To design, build and test a low-cost, battery-powered prototype that warns both the machine side and the worker side when a person or obstacle comes within a hazardous distance of a mining machine, using two independent sensing methods connected by a router-free wireless link, so that the failure of one method or link does not remove the warning.

3.3 The Idea

A man working next to a large truck. For the truck to know there is a worker nearby and vice versa, the truck and the worker will be equipped with two different types of detection systems.

1. The truck's own eyes. Sensors in the truck. There are a radar and a laser rangefinder in the truck. They scan ahead for obstacles. When one detects an object close enough, an alarm is triggered, the camera makes a snapshot, and it alerts the worker by wireless radio.

2. Pair of radio tags. A worker wears a UWB anchor and a truck has a UWB tag. They measure the distance between each other by measuring the time it takes to send radio pulses. When distance becomes too small, an alarm in the worker's unit sounds its alarm and radios the news to the truck.
Either one works well alone. Together, however, they complement each other because the radar can detect a person who is not wearing the tag, and vice versa – the tag can "see" the worker hiding around the corner where the radar cannot have the line of sight.

3.4 Scope and limits

The prototype shows the proof of concept on bench and small scale field tests. The threshold for the alarm in the algorithm is set to 1.5 m. Such a distance would be good for the bench test but not sufficient for a loaded dump truck, which has a much larger stopping distance, and therefore the thresholds are specified as adjustable parameters. This system detects people but cannot slow down or stop the machine. The camera takes pictures but does not identify people.

4. Objective of the Project

The overall aim was to construct a functional, redundant proximity warning system prototype for mining machinery and personnel. The objectives have been specified as follows.

1. Detection of the presence of a human, even a barely moving person, through a 24 GHz millimetre wave radar, and alarm when the person is detected at a distance of less than 1.5 m.

2. Detection of any solid object within 1.5 m by a laser distance sensor based on time-of flight measurement.

3. Generation of a local warning sound and light alarm by a buzzer and LED even in the event of failure of all wireless links.

4. Displaying of only relevant information (alert/clear, human/object detected, distances, and status of links) on an OLED screen.

5. Capturing a photo each time the system generates an alarm without recording videos in order to conserve energy.

6. Monitoring of the distance between a miner and a machine by UWB distance measurement, and alarming the miner when the distance falls below a predetermined threshold value.

7. Interconnection of the machine unit, the camera unit, and the worker unit via ESP-NOW in order to operate the system as a whole without the aid of a router and internet.

8. Make the two detection paths redundant: a failure in one sensor, one radio link or one node must not silence the warning from the other path.

9. Run the machine unit from a rechargeable 18650 battery, and document the power budget.

5. Literature Review/Background

Background

Mine workings are one of the most dangerous working environments because of low visibility, confined space, dust, and continuous movement of the machines like loaders, dumpers, and trucks. In such an environment, the machine operator may find it difficult to see the workers and obstacles around him, especially in the blind spot areas. Any late detection of people or obstacles around him might result in severe accidents, injuries, damages, and operational delays. Traditional safety mechanisms like wearing reflective clothing and manual monitoring might not always be sufficient in a tough underground environment.

To solve this issue, the project aims at developing a **cost-effective multi-sensor collision avoidance system for underground mines**. The proposed system uses the concept of detecting human presence using radar, measuring the distance between the object and human, and taking wireless communication along with capturing images using cameras for a certain range.

The system comprises of :

a. Proximity detection in mining

The initial version of approved proximity detection systems for use in underground coal mines uses electromagnetic fields. The machine creates a magnetic field, and the sensor in miner's wear detects it. NIOSH, the US research institute working on occupational safety, listed the disadvantages: the need to equip the miner with an additional component, environment and machine interference effect, and limitations of the location range and precision [3] As a result, NIOSH studied other options like RFID, LIDAR, radar, ultrasonic detection, computer vision, and the possibility of using sensors in combination [3][4]. Later, NIOSH researched radar and LIDAR and identified variables affecting their detection performance [4]

In surface mining, NIOSH studied commercial radar and tag-based proximity systems for haul trucks. These results showed the complementarity of radar and cameras and stated that the use of radar systems should be combined with camera systems and not replace them [5]]. It is similar to the structure of this project: radar as detection, camera as evidence, and tag-based system as a back-up.

b. Millimetre-wave radar

The radar emits radio waves and receives the reflection. Millimeter-wave (mmWave) radars transmit at very high frequencies, hence making the antenna and electronic components to be of smaller sizes that can be placed in the board having a diameter of a coin. In this case, the DFRobot C4001 transmits at 24 GHz using the frequency modulated continuous wave (FMCW) modulation and thus can detect distance and speed. The 12 m model has a detection angle of 100 degrees, presence detection range of 8 m, motion and ranging detection range of 12 m and

a speed range of 0.1 to 10 m/s [14]. Another 25 m model is available [14] Unlike the infrared sensor, the radar works even under total darkness conditions and does not depend on the color and light.

c. **Time-of-flight laser distance sensing**

The ST VL53L0X is a compact ToF sensor. This uses a 940nm invisible laser source and single photon detector array. It is able to detect absolute distances from 0 to 2m. Importantly, it does not matter what colour the target object is [15]. The sensor has an accuracy of ±3% and measures distances in less than 30ms whilst consuming only around 20mW [16]. It can be relied upon to measure distances of around 1.2m in typical room environments. However, it can be confused by bright sunlight, reflective surfaces and transparent objects. Because of this, it was decided to use it as a secondary simple sensor.

d. Ultra-wideband ranging

Ultra-wideband (UWB) radio systems transmit very short pulses using a wide frequency range. An ultra-wideband physical layer has been defined within the IEEE 802.15.4 standard, with an improved ranging integrity defined within the 802.15.4z amendment [10]. Pulses transmitted at a bandwidth of 500MHz have pulse widths of a fraction of a nano second; meaning that accurate timings can be achieved even in high reflecting environments [11].

Two-way ranging is used to determine the distance between two devices. One device will transmit a request whilst the other will reply, this time taken to make the round trip will be divided by 2 and then multiplied by the speed of light to give an accurate distance. Ultra-wideband can provide accurate results down to centimetre level, usually with accuracies of around 10cm for distances of up to 200m [12]. Whereas signal strengths do not provide accurate distance readings due to the variation [11].

The transceiver used in this module is an ultra-wideband UART interface controlled by simple AT commands. The transceiver model used is the Reyax RYUW122 (Lite), which states an accuracy of around 10cm in an open field environment [13]. This accuracy will be given for open spaces however when indoors, close to metal or when there is a body between the antennas results can be more variable, as discussed within section 6.

e. **ESP-NOW ESP-NOW**

ESP-NOW is a connectionless protocol defined by Espressif. This allows for small messages to be sent between ESP32 boards. These messages have a maximum size of 250 bytes, and a default speed of 1 Mbps [17]. Broadcast addresses can be used which means that no pairing is required. However, there is no encryption available for broadcast packets. There is also no confirmation that these packets have arrived. For this reason, the status messages are sent multiple times per second [17].

f. Related work and the gap this project fills

UWB proximity warning system has been proposed for construction sites, using a wearable unit and a zone unit to simultaneously alert workers [18]. Work undertaken by NIOSH has shown similar results using radar, LIDAR and sensor fusion in mining [3][4][5]. Commercial radar based proximity warning systems for trucks are becoming available on the Indian mining market [9]. This project has been completed with a student focus. Therefore, a compact and fully documented proximity warning system combining radar, laser sensor, camera and UWB ranging has been created. The system provides bidirectional status messaging between the machine and worker. Furthermore, it has been intentionally designed so that both alert paths will still provide an alarm if the other is loss.

5.1 Comparative Analysis of Existing Technologies

The major technologies discussed in the literature can be summarized as follows:

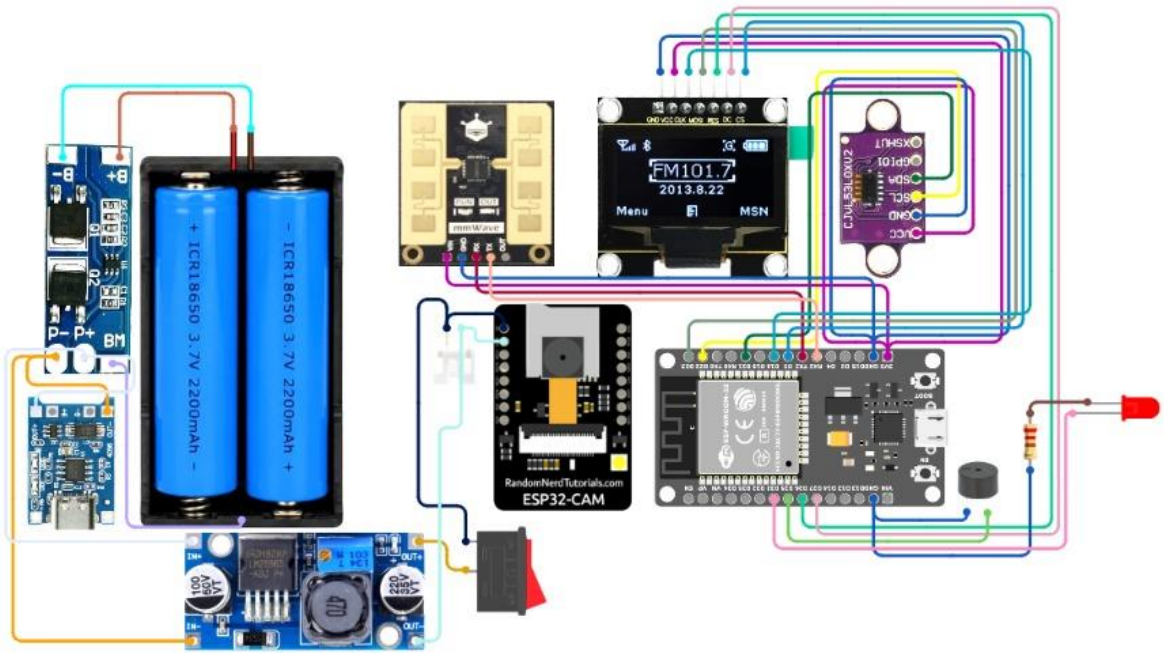
Technology	Sensors/Method	Major Strength	Major Limitation
Proximity Detection System	RFID, UWB, Magnetic Field	Works in dark and dust	Wearable tags and false alarms
Collision Avoidance System	UWB + Radar + Logic	Can provide automatic warning/stop	High cost and difficult integration
Radar Detection	mmWave / FMCW Radar	Works in dust, smoke and low light	Poor object classification
LiDAR / 3D Mapping	2D/3D LiDAR	Accurate distance and shape	Dust sensitivity and high processing
Camera + Computer Vision	RGB Camera + AI	Good object classification	Poor performance in dust and low light
Thermal Imaging	Infrared Camera	Detects humans in darkness	Hot machine parts can cause confusion
Ultrasonic Sensors	Ultrasonic	Low cost and simple	Short range and false alerts
NIOSH Prototypes	Multi-sensor testbeds	Scientifically validated	Limited scalability
Fleet Tracking / Geofencing	RTLS + Maps	Traffic and zone control	Not sufficient alone for real-time collision prevention

6. Methodology

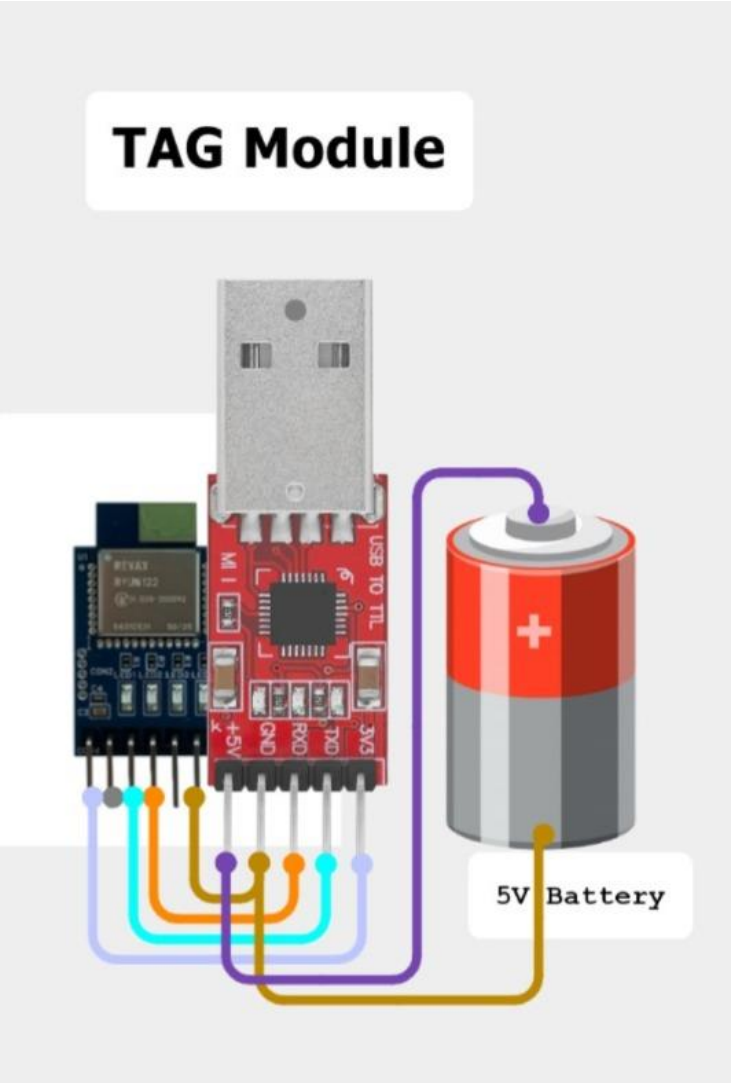
6.1 Approach

We followed a build-and-test-in-layers approach. Each subsystem was made to work on its own on the bench before the next was added, and a working sketch was never rewritten without a good reason. This turned out to matter: the one time a validated radar routine was replaced by a new one, the behaviour changed and had to be restored (Section 7).

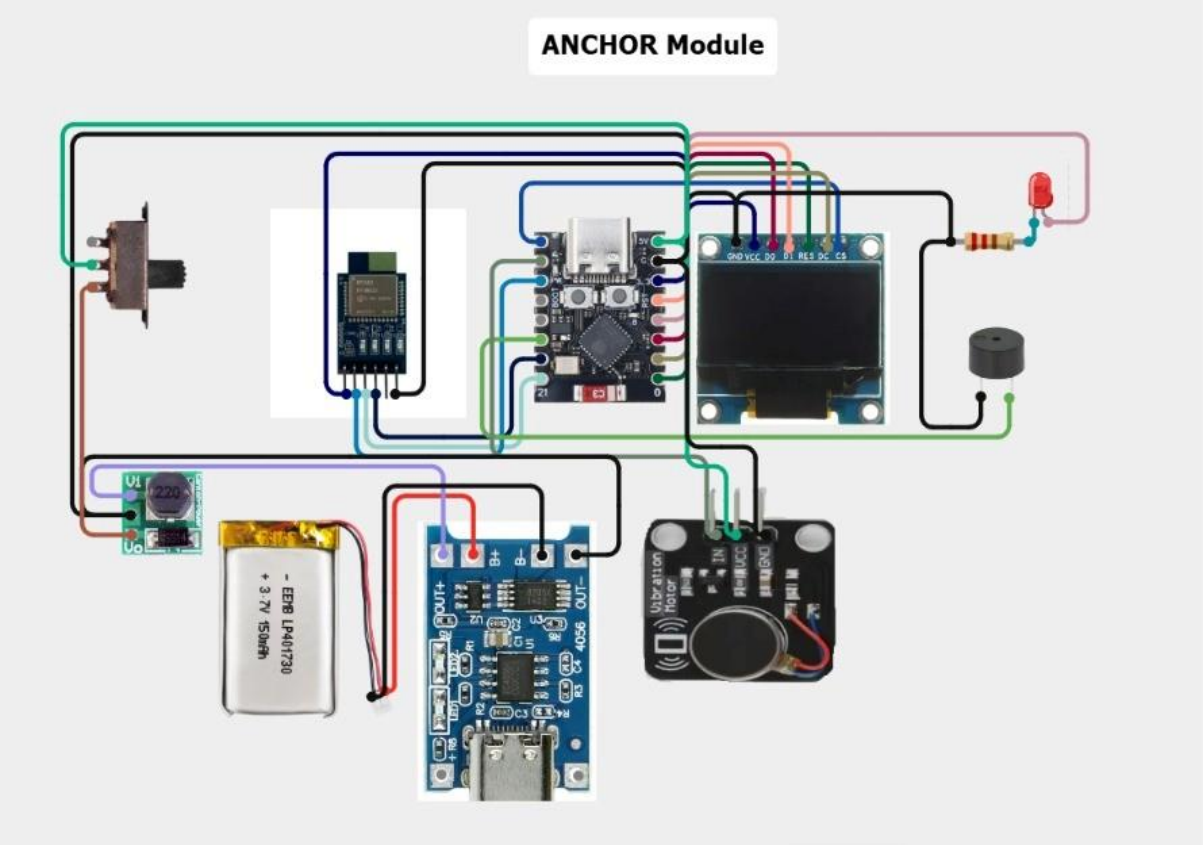
6.2 System architecture



6.1 Circuit Diagram of Module 1 – Machine Node (mmWave Radar + ToF Sensor)



6.2 Circuit Diagram of UWB Tag Module (Machine Node)



6.3 Circuit Diagram of Module 2 – Anchor Module (Worker Node)

6.3 Key design decisions

Decision	Reason
Two independent detection paths	A radar sees people who carry no tag. A tag sees workers outside the radar's line of sight. Each covers the other's weakness
Local alarm before any radio work	The buzzer and LED switch on first in the program, so they cannot be delayed by the wireless link
ESP-NOW broadcast on a fixed channel	No router, no pairing, no MAC addresses to type in. A magic first byte marks the project's packets and others are ignored
Each node sends only its own detection	If a node also forwarded the other node's alarm, the two could keep each other's alarm latched. Sending only own detection avoids that loop
Repeated status packets	Broadcast has no acknowledgement, so a lost packet is simply replaced by the next one within 100 to 200 ms
Median filter, hysteresis and confirmation on UWB distance	UWB readings scatter, and a plain "distance below limit" test would make the alarm flicker at the edge
Camera snapshots only on alert	Saves power, and every alert leaves a record

6.4 Alarm logic

Node	Buzzer and LED turn on when
Machine node	the radar sees a person within 1.5 m, **or** the laser sees an obstacle within 1.5 m (held for 2 s), **or** the worker unit reports that the worker is inside the UWB alarm distance
Worker node	the UWB distance is below the alarm distance (with hysteresis), **or** the machine reports a human or obstacle in front

If the ESP-NOW link drops, each node still alarms from its own sensors. If the UWB path fails, the radar and laser still alarm and tell the worker. If the radar path fails, the UWB distance still alarms on both sides.

6.5 Work plan

Phase	Activity	Outcome
1	Radar, laser sensor, OLED, buzzer and LED on the ESP32 sensor node	Local human and obstacle alarm inside 1.5 m
2	ESP32-CAM with SD card and the first ESP-NOW link	Photo saved on each alert; "CAM:OK" shown on the OLED
3	Battery, charger and boost converter; power calculation	Portable operation and runtime estimate
4	UWB module bring-up using the ESP32 as a serial bridge, anchor and tag configuration, distance readout	Working distance measurement between anchor and tag
5	Distance filtering, alarm logic and OLED on the ESP32-C3 worker unit	Stable worker-side alarm
6	Two-way ESP-NOW integration of machine and worker nodes	Firmware for redundant alarms (field test pending)
7	Testing, documentation and this report	Final report

7. Work carried out during the reporting period

7.1 Machine Node (Radar, Laser Sensor, Display & Alarm)

The machine node is based off the ESP32 board. The radar (C4001) connects to the board via UART (9600 baud rate) using pins GPIO16 & 17. The laser sensor (VL53L0X) connects via I2C using pins GPIO21 & 22. It also uses pin GPIO23 for resetting the module. The display uses SPI connecting via pins GPIO18, 19, 25, 26 & 27 to the 1.3inch SH1106 OLED. The Buzzer uses GPIO14 and Red LED uses GPIO33.

The radar is in speed mode. This means that it will output the distance to a target in metres allowing the program to decide if the distance is greater than 1.5m. The radar has been set to detect objects between 30cm-300cm with a threshold of 10. Fretting detection is on to ensure almost stationary people are picked up. It also gives speed & signal energy. These can be seen on the serial monitor whilst adjusting the settings. There is another mode called presence. However, speed mode was used as it allows for values to be shown and adjusted.

The laser sensor reads every 100ms. Long range mode has been selected to allow it to read up to approximately 1.5m. Any readings that are invalid or over 2m are ignored. If a valid reading is less than 1500mm then an obstacle is detected. If there is a trigger from either the radar or laser within 1.5m the buzzer and LED will turn on. The alarm will stay on for 2 seconds after the last detection. This prevents the alarm from flickering if someone is stood exactly on the edge. The screen will show HUMAN if it was triggered by the radar or OBSTACLE. The display only shows essential information. This includes the camera connection status, CLEAR or Alert Box, Laser Distance, Radar Distance & Event Count.

7.2 Camera Node

The camera node uses the AI-Thinker ESP32-CAM. It awaits the status messages from the machine node using ESP-NOW. Once an alert has been triggered and 3 seconds or more have passed since the last picture has been taken, a VGA JPEG will be taken. This will then be saved using the next available filename starting with alert_ (alert_1.jpg, alert_2.jpg etc). This means that if the device is restarted previous images won't be overwritten. The red LED will illuminate whilst an alert is being triggered. Every second a heartbeat message will be sent to ensure the OLED on the machine shows CAM:OK. If no status messages are received for 1 second then it is assumed there isn't an alert and false images won't be taken. The memory card is set to 1-bit mode to leave the flash LED pin available. However, the flash wasn't used in the prototype to save power.

7.3 Power System

The machine unit uses 1 x 18650 Li-ion cell. This is charged using a protected TP4056 module. A boost converter converts the voltage from 3.7v to 5v. This powers the ESP32, ESP32-CAM & Radar. A 470-1000 Microfarad capacitor is placed across the ESP32-CAM to help with the current spike when turning the camera on. The estimated current usage at 5V is as follows.

These are typical values from datasheets.

Part	Typical current	Peak current
ESP32 DevKit (with ESP-NOW)	100 mA	250 mA
ESP32-CAM (flash off)	150 mA	310 mA
C4001 radar (assumed)	100 mA	100 mA
VL53L0X	20 mA	20 mA
OLED	15 mA	25 mA
Buzzer (average)	10 mA	30 mA
LED	1 mA	4 mA
Total at 5 V	**about 400 mA**	**about 740 mA**

That is about 2.0 W typical and 3.7 W at peak (P = V x I). With a boost converter efficiency of 85 percent and a battery voltage of 3.7 V, the battery current is (5 x 0.40) / (0.85 x 3.7), or about 0.63 A typical and about 1.2 A at peak. A 2200 mAh cell gives roughly 2000 mAh of usable capacity, so the estimated runtime is about 3.2 hours typical and about 1.7 hours in the worst case. Two cells in parallel double those times. The runtime must be measured on the real hardware, and the worker unit's consumption still has to be measured separately.

Power system of the worker node

The worker unit is worn on the arm, so it has to carry its own battery. It is built from an ESP32-C3 development module, the RYUW122 UWB module (anchor), a 1.3 inch OLED, a vibration module, a buzzer, a red LED, a LiPo battery, a TP4056 charging module, a boost converter and a power switch.

Power path. The LiPo cell feeds the TP4056 module, which charges it and protects it. The cell's output goes through the power switch to the boost converter, which raises it to 5 V. That 5 V goes to the 5 V pin of the ESP32-C3 board, whose onboard regulator makes 3.3 V for the ESP32-C3, the UWB module and the OLED. The buzzer and the vibration module are driven from the 5 V rail. Because the 3.3 V regulator is linear, a 3.3 V load draws the same current from the 5 V rail as it does at 3.3 V, so every load below is counted at 5 V.

Current budget (milliamps at the 5 V rail):

Part	Tracking (typical)	Alarm active (typical)	Peak
ESP32-C3 module, Wi-Fi radio on for ESP-NOW	85	85	300
RYUW122 UWB anchor, ranging continuously	80	80	181
1.3 inch SPI OLED	15	15	25
Active buzzer (beeping at 50 percent duty)	0	15	30
Red LED with resistor (blinking)	0	3	6
Vibration module (running)	0	80	120
Total	**180 mA**	**278 mA**	**662 mA**

Where the figures come from. The Reyax datasheet lists the UWB module at about 8 mA in anchor mode, 86 mA while transmitting and 181 mA while receiving [13]. The anchor sends a request and waits for the reply many times per second, so its average is much higher than the 8 mA idle figure. I assumed 80 mA on average and used the 181 mA receive current as the peak. For the ESP32-C3, the Espressif datasheet gives a receive current of roughly 84 mA and transmit peaks of up to about 335 mA at full duty cycle

(https://documentation.espressif.com/api/resource/doc/file/aYwvqd1B/FILE/esp32-c3_datasheet_en.pdf).

The board listens all the time for ESP-NOW messages and transmits only short bursts, so I used 85 mA typical and 300 mA peak. The OLED, buzzer, LED and vibration module values are typical figures for these parts and are estimates, not measurements.

Power and battery current. At 5 V (P = V x I), the load is 0.90 W while tracking, 1.39 W with the alarm on and 3.31 W at peak. Assuming a boost converter efficiency of 85 percent and a nominal cell voltage of 3.7 V, the battery current is I_bat = (5 V x I_load) / (0.85 x 3.7 V):

- Tracking: (5 x 0.180) / (0.85 x 3.7) = about 0.29 A
- Alarm active: (5 x 0.278) / (0.85 x 3.7) = about 0.44 A
- Peak: (5 x 0.662) / (0.85 x 3.7) = about 1.05 A

Estimated runtime. Only about 80 percent of a LiPo cell's rated capacity is usable before the protection circuit cuts off. The mixed column assumes the alarm is on for 10 percent of the time (average battery current about 0.30 A). The peak column is a worst case that will not last in practice.

LiPo capacity	Usable (80 percent)	Tracking only	Alarm on all the time	Mixed (90 percent tracking, 10 percent alarm)	All loads at peak
500 mAh	400 mAh	1.4 h	0.9 h	1.3 h	0.4 h
1000 mAh	800 mAh	2.8 h	1.8 h	2.7 h	0.8 h
2000 mAh	1600 mAh	5.6 h	3.6 h	5.3 h	1.5 h
3000 mAh	2400 mAh	8.4 h	5.4 h	8.0 h	2.3 h

For a full 8-hour shift the worker unit needs about 2.4 Ah of usable capacity at 0.30 A average, so a LiPo cell of about 3000 mAh is the sensible choice, with a little more for margin. A 1000 mAh cell gives under 3 hours, which suits a demonstration but not a shift. The vibration module is not working in the current prototype; without it the alarm-on current drops to about 198 mA, or 0.31 A from the battery.

Notes on the other parts.

- **TP4056 charger:** the default charge current is 1 A, which suits cells of about 1000 mAh and above. For a smaller cell, raise the programming resistor to bring the charge current down to 0.5 C to 1 C of the capacity. Use the version with the protection chip so the cell is protected against over-discharge and short circuits.
- **Boost converter:** it must deliver at least 0.7 A at 5 V and accept more than 1.1 A at its input, to cover the peak case.
- **Power switch:** rate it for at least 2 A and place it in the battery positive line, so that the unit can be turned off while the charger stays connected.
- **Not included in the numbers:** the quiescent current of the boost converter, the USB chip and power LED on the development board (a few milliamps), cell self-discharge, and the fall in boost efficiency as the cell drops toward 3.0 V. The real runtime will therefore be somewhat shorter than the table.
- **Ways to extend the runtime:** range every 200 to 300 ms instead of every 100 ms, dim or switch off the OLED when there is no alarm, pulse the vibration module instead of running it continuously, and remove the development board's power LED.

As with the machine node, these are design estimates. The worker unit's real consumption should be measured with a USB power meter or a multimeter in series with the battery, and the table updated with the measured values.

7.4 UWB Ranging Subsystem

Each Reyax RYUW122 Lite module is controlled using AT commands over UART. As there was no USB to TTL converter available initially, one of the ESP32s had to be used as a transparent serial bridge. The NRST pin on the module was found to be useful to connect to a GPIO pin for resetting the module on programme startup.

The modules were configured with a common network ID REYAX123, common password and different addresses. One module was set to be the Anchor (mode 1), whilst the other was set to be the Tag (mode 0). The addresses used were ANCHOR01 and TAG00001 respectively. The address has to be exactly 8 characters long, therefore initially I chose shorter addresses ANCHOR1 and TAG1. This caused the initial ranging requests to fail, returning nothing. See section 7.

Once the module was configured, ranging could be achieved simply by sending a command to the anchor specifying the tag. A line of text will be returned containing the address of the tag, payload and distance in cm between the devices. The firmware will extract the distance from this. As the UWB readings can be quite inconsistent, the following steps have been implemented:

1. **Median of the past 5 readings**, so isolated spikes can be ignored.
2. **Hysteresis:** of the alarm. Only triggers when below 150cm, does not trigger until above 170cm.
3. **Confirmation:** Two consecutive readings must be under the limit before triggering an alarm.

If the tag fails to respond for 5 consecutive polls, then there will be no signal detected and the history will be reset.

7.5 Worker Unit

The worker unit is built around the ESP32-C3 Super Mini. The UWB Anchor is connected over UART1, with TXD connected to GPIO20, RXD connected to GPIO21 and reset connected to GPIO7 (baud rate 115200). The 7 pin SH1106 OLED is connected using soft SPI to pins GPIO2, 4, 0, 1, 5. This was found to be more reliable than hard SPI on this board. (See section 7). The buzzer is connected to GPIO6, the red LED to GPIO3 and vibration motor module to GPIO10. The OLED displays the distance to the machine in large text, the front status (HUMAN, OBJ with distance or clear) and link status.

7.6 Bi-directional ESP-NOW integration

All nodes are using Wi-Fi channel 1. Two types of messages are sent:

- **Machine to Camera and Worker:** This is the original 10 byte status message the sensor node was sending. It begins with the magic byte 0xA5. It contains the alert status, flags byte (bit 0 = Human from the radar, bit 1 = Obstacle from laser), distance in mm from the laser and counter. This message has not changed as the camera node is already working.

- **Worker to Machine:** This is the new 7 byte message with the magic byte 0xA7. It contains a flag stating whether a valid UWB reading was received, the distance in cm from the UWB, the workers own hazard flag and a counter. Since this message has a different magic byte and length, the camera node ignores it.

The machine node has been extended. The radar, laser, display and camera link code are the exact same as the validated sketch. New lines have been added to receive the message from the worker, display the distance from the worker (W:120cm) on the OLED and add the banner WORKER NEAR!. The buzzer and LED now trigger on own alert OR worker too close. The worker node receives the message from the machine to display the front status. If the machine reports a human or obstacle then an alarm will trigger. Both nodes will report a lost link after 2 seconds without receiving a message. By default, this will just change the status to LOST but can be adjusted to play a warning chirp.

8. Results and Discussion

8.1 Status of each subsystem

Subsystem	What was done	Status
Radar human detection	C4001 in speed mode, window 30 to 300 cm, alarm gated at 1.5 m	Working on the bench: people are detected across the window and the alarm starts inside 1.5 m
Laser obstacle detection	VL53L0X in long-range mode, alarm below 1500 mm	Implemented in the validated sensor-node sketch; distance shown on the OLED
OLED status display	SH1106, essential information only	Working on both nodes
Local alarm (buzzer and LED)	Switches on before any radio work	Working on both nodes
Camera node	Snapshot to SD on alert, heartbeat to the machine node	Implemented and used with the machine node
UWB ranging	Anchor and tag, 8-character addresses, distance parsed and shown	Working; readings scatter more than the datasheet figure (see 6.3)
UWB distance filtering and alarm	Median of 5, hysteresis 150/170 cm, 2-reading confirmation	Working on the worker unit
Vibration motor	Motor module driven from the worker unit	Not working; parked (see Section 7)
Two-way machine and worker link	Extra packet type, shared alarm rules	Firmware written; integrated field test pending

The measured values in this report are the ones the bench work produced. Where a figure is an estimate (for example power and runtime), it is labelled as such.

8.2 Radar and Obstacle detection

The initial implementation of the radar used presence mode. This didn’t provide the desired effect. Switching to speed mode, using fretting detection, and a software based range gate of 1.5m provided the desired function. The radar can detect people across the detection window but will only trigger an alarm if the closest object is within 1.5m. Having a larger detection range means that a person can be seen on the serial monitor, therefore making the threshold tuning much easier. The 2s delay prevents the alarm from chattering on the edge. The laser also provides a simple secondary confirmation of solid objects at short ranges, especially static ones where there is little data available to the radar.

8.3 UWB ranging: what the readings look like

The UWB link worked as soon as the module settings were correctly set. It was possible to read and display the distance between the anchor and tag in cm. However, the readings weren't as consistent as hoped. The vendor say it should be accurate to 10cm in an open field [13]. The technology can provide cm distance measurement [12]. However a bench in a lab isn't an open field. There are many factors which are known to cause problems with short range UWB distance measurements:

- Reflection off walls, tables, floor and metal objects.
- The human body blocking or detuning the antenna – very important on a wrist worn device.
- Antenna alignment – Antennas should face each other.
- Electronic devices and wiring close to the antenna – A busy breadboard.
- A constant offset that has not been calibrated out.

This was addressed in the firmware code by seeing the distance value as a noisy data value. Using a median filter removes individual spikes. The hysteresis band (150cm on, 170cm off) ensures the alarm doesn't toggle when close to the threshold. Requiring 2 consecutive low readings means that the alarm doesn't react if there is a single bad reading. This does add some delay to the reaction time (2-3 readings). However, this is acceptable as the person will be walking towards the device. Section 6.7 shows a separate test has been planned to fully characterise the error in the readings. This involves taking 100 readings at several known distances. The average offset will then be known. The module can also be put into a calibration mode to remove any fixed offset.

8.4 Behaviour under failure (design intent)

As the main claim of this project is redundancy the following table shows how the system should behave if there is a failure. This follows directly from the alarm rules in section 4.4. This will need to be verified in the final integrated field test.

What fails	Expected behaviour
Camera node off or broken	Local alarms on both nodes keep working; the machine OLED shows CAM:--
ESP-NOW link between machine and worker lost	Each node alarms from its own sensors; each OLED shows link LOST
UWB tag or anchor not answering	The worker unit shows no signal; the radar and laser still alarm on the machine and the machine still tells the worker
Radar fails to start	The machine OLED reports it and the program waits, so a missing safety sensor is never hidden
Laser sensor missing	The radar continues to work
One ESP-NOW packet lost	The next status packet arrives within 100 to 200 ms
Worker unit switched off	The machine still alarms from its own radar and laser

8.5 Why the distance chosen is 1.5m for a bench setup

By doing a simple calculation you can see that a longer distance needs to be used for real vehicles. A truck going at 20km/h travels around 5.6m every second. When the alarm is raised and there is a 1.5 second reaction time by the driver they will have travelled a distance of about 8.3m before they start braking. So having an alarm at just 1.5m would be far too late. The 1.5m limit was used in the prototype as it is safe and easy to test. All distances have names and can be increased, or changed to be dependant on the speed of the machine in a future version. This calculation is not an actual measured stopping distance.

8.6 Limitations

- The system will give an alert but will not stop the machine.
- UWB range finding has a higher scattering distance than the best case scenario given in the datasheet. This will also vary depending on the environment.
- The radar is more sensitive to objects moving towards/away from it than to those moving laterally. It cannot distinguish between a person and another moving object.
- The laser sensor is only reliable up to around 1.2m. It is also affected by bright sunlight, reflective and transparent surfaces.
- The ESP-NOW broadcast messages aren't acknowledged or encrypted. This means that there is a possibility that a fake packet could be sent to cause or prevent an alarm.
- The camera doesn't have any night lighting in this prototype meaning pictures taken in the dark will be dark.
- The radio range hasn't been tested underground, around corners or through rock. The housing, batteries and electronics aren't rated for hazardous atmospheres. Nothing has been approved by a mining regulator.

8.7 Test protocol for the measured results

The table below is the protocol for the integrated field test. The observed column is left empty on purpose, so that real measurements can be entered and kept in this report.

No.	Test	What to record	Observed
T1	Radar alarm distance: person walks toward the machine node	Distance at which the alarm starts, five repetitions	
T2	Laser alarm: flat object moved toward the sensor	Distance at which the alarm starts	
T3	UWB accuracy: modules at 50, 100, 150, 200 and 300 cm	100 readings each, mean and spread	
T4	UWB alarm: worker walks toward the machine	Distance at alarm start and end, compared with the 150/170 cm settings	
T5	ESP-NOW link range and loss	Maximum distance with CAM:OK and link OK; packets lost	
T6	Camera: alert triggers a photograph	Photo present on the SD card, delay from alarm	
T7	Redundancy: switch off the worker unit, then the tag, then the camera node	Does each remaining path still alarm	
T8	Battery run time	Hours until shutdown, compared with the 3.2 h estimate	

9. Challenges Encountered and Mitigation Measures

Challenge	What happened	How it was handled	Status
No USB-to-TTL converter	The UWB modules could not be configured directly from a PC	An ESP32 was programmed as a transparent serial bridge; every command had to end with a carriage return and line feed	Solved
UWB module damage	The chip burned and received nothing at any baud rate on several pin pairs	Bought a new module, Checked wiring, scanned baud rates and pin orientations, wired the NRST reset pin, and noted that GPIO16 and GPIO17 are tied to PSRAM on WROVER boards	Solved
Wrong address length	Ranging returned nothing. The module expects addresses of exactly 8 characters, and I had used ANCHOR1 and TAG1	Changed to ANCHOR01 and TAG00001 on both modules and checked the settings by reading them back	Solved
Scatter in UWB readings	Distances jumped from reading to reading	Median filter, hysteresis and confirmation logic; calibration and an antenna placement study planned	Mitigated
OLED showed noise or stayed blank	Driver-chip uncertainty and the default hardware SPI pins on the new board	Tested SH1106 and SSD1306 drivers with a small program and moved to software SPI, which works on any pins	Solved
Change of microcontroller	The worker unit moved from an ESP32 WROVER to an ESP32-C3 Super Mini, so all pins changed. GPIO18 and GPIO19 are the C3's USB lines, and a buzzer wired to GPIO20 collided with the UWB receive pin	Re-planned the pin map, kept the UWB on GPIO20 and GPIO21, and moved the buzzer to a free pin	Solved
Radar behaviour changed after a code rewrite	A new radar routine written from a library example did not behave like the validated one: alarm distance and settings differed	Went back to the validated sensor-node sketch and added the worker link on top, with the new lines clearly marked	Solved
Keeping the camera node untouched	Adding a new message type could have broken the working camera node	The worker message uses a different magic byte and a different length, so the camera node ignores it	Solved

Library differences	The ESP-NOW receive callback has a different form in older and newer Arduino ESP32 core versions	The code selects the correct form at compile time	Solved
Broadcast limits	ESP-NOW broadcast has no acknowledgement and no encryption	Status messages repeat every 100 to 200 ms and carry a magic byte; encrypted links are listed as future work	Mitigated
Vibration motor does not run	The 3-pin motor module (IN, VCC, GND) did not vibrate on any pin, even after replacing it	Parked. The worker unit alarms through its buzzer and LED. Next step: test the module with its input tied to 5 V and to ground, and try a transistor driver	Open

10. Deliverables Achieved

- A working machine node that detects humans (radar) and obstacles (laser) and alarms inside 1.5 m with a buzzer, an LED and an OLED display.
- A camera node that saves a photograph to an SD card on every alert and reports its health to the machine node.
- A battery power system (18650, TP4056, 5 V boost converter) with a documented power budget and runtime estimate.
- A working UWB ranging link between an anchor and a tag, with filtered distance and an alarm threshold.
- A worker unit on an ESP32-C3 Super Mini that shows the distance to the machine and alarms.
- Firmware for the two-way ESP-NOW link between the machine and the worker unit, with shared alarm rules and link supervision.
- Documentation: the detailed explanatory report, pin tables, packet formats and this final report.

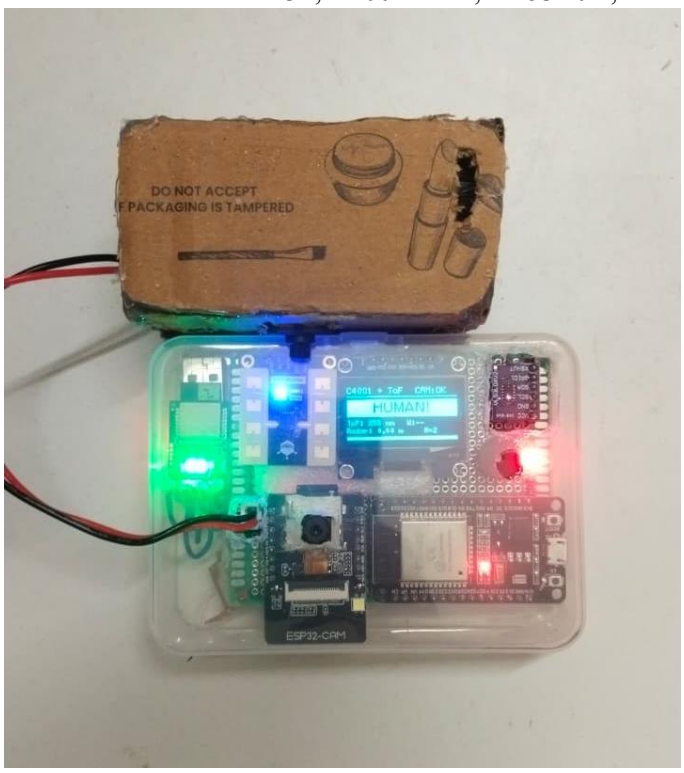
11. Publications, Patents, Prototypes, or Software Developed

Publications: none so far. The redundant two-path design, with a camera-backed record and a lost-link strategy, could form the basis of a short conference or workshop paper once measured field data are available.

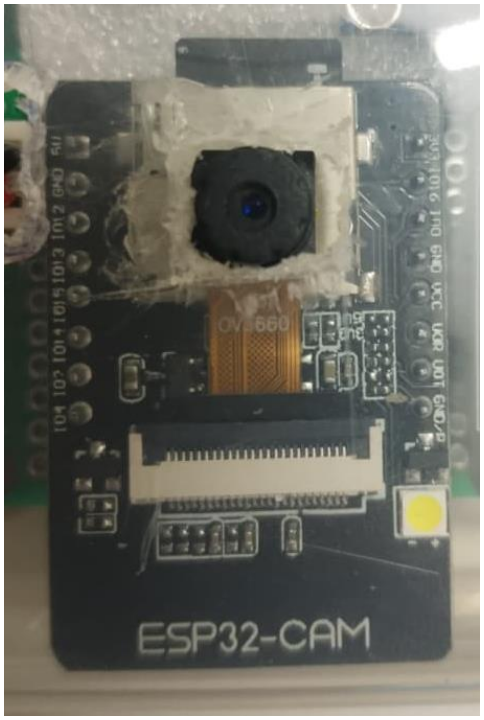
Patents: none filed.

Prototypes:

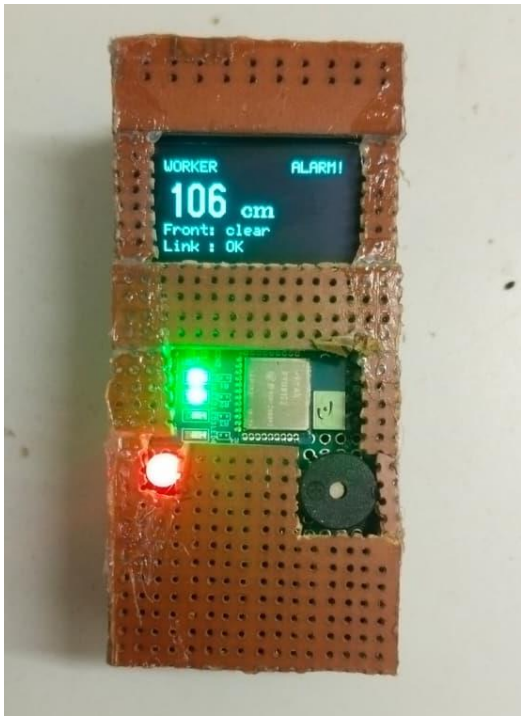
1. Machine node: ESP32, C4001 radar, VL53L0X, OLED, buzzer and LED.



2. Camera node: ESP32-CAM with an SD card.



3. Worker node: ESP32-C3 Super Mini, RYUW122 anchor, OLED, buzzer, LED and motor module.



4. UWB tag unit on the machine side (power only).



5. Battery pack: 18650, TP4056 and boost converter.

Software (Arduino IDE, ESP32 board package):

- machine_node: sensor node firmware extended with the worker link.
- worker_node: UWB ranging, filtering, display, alarm and ESP-NOW link.
- cam_node: camera snapshot firmware with heartbeat.
- Libraries used: DFRobot_C4001, Adafruit_VL53L0X, U8g2, esp_now, esp_wifi, esp_camera and SD_MMC.

12. Future Work / Recommendations

1) Test the system in the field. The next major step is to carry out the field test planned in Section 6.7. We need to record real-world measurements—specifically the UWB distance errors and the ESP-NOW range—to see how the system actually performs outside the lab.

2) Fine-tune the UWB and antenna placement. We should use the UWB module's built-in calibration tools to fine-tune the distance readings. Additionally, we need to experiment with the best physical locations to mount the antennas on both the worker's wrist and the machine to ensure a reliable signal.

3) Adjust the warning distances for real stopping times. The current warning thresholds need to be increased to match the actual stopping distance of the machinery. Ideally, the system should be smart enough to automatically increase the alert distance if the machine is moving faster.

4) Connect directly to the machine's controls. A warning is only useful if the operator has time to react. To make the system truly effective, it should be wired into the machine's controls—for example, using a relay to automatically cut traction or sound an alarm inside the cab.

5) Track multiple workers. To make the system usable for a full crew, each tag needs a unique address so the base station can track several people at once. The display could then show either the distance to the nearest worker, or a count of how many people are currently in the danger zone.

6) Secure the wireless connection. We need to make the ESP-NOW link more secure by using unicast and encryption (or an authenticated message format). Adding an acknowledgement feature would also ensure that alarm messages are received and not lost.

7) Improve camera visibility in the dark. To help the camera work at night, we should add an infrared lamp. For extremely dark or dusty conditions where normal cameras struggle, integrating a thermal sensor would be a major improvement.

8) Fix the vibration motor. The vibration motor needs to be repaired or replaced. In a noisy mine, a physical vibration is a much better way to alert a worker than a buzzer, which might not be heard over the machinery.

9) Add battery monitoring. The worker unit needs power management features. We should add low-battery warnings to the OLED screen and measure the actual power consumption to make sure the battery can easily last a full work shift.

10) Make the design rugged and compliant. The final device needs to be housed in a tough, durable enclosure. Most importantly, it must be designed to meet the strict safety rules for hazardous atmospheres required by the specific mine where it will be used.

11) Expand the radio options. We need to measure how far the radios actually transmit underground. If the range isn't sufficient, we should explore alternative options like LoRa, which offers longer transmission distances.

12) Ensure regulatory compliance. Before this system is deployed in the real world, the design must be thoroughly checked against the safety requirements of local regulators, such as DGMS in India or MSHA in the US.

13. Conclusion

This project set out to prove that a small, low-cost system can successfully monitor the area around a mining machine and alert both the operator and the worker even if part of the system fails. The machine node uses a millimetre-wave radar and a laser distance sensor to spot people and obstacles within 1.5 meters. When it does, it triggers a local buzzer and LED alarm , and uses a camera to record the event. Meanwhile, a UWB anchor and tag track the distance between the worker and the machine, with the worker's unit warning them if they get too close. To keep things simple, ESP-NOW connects the nodes without needing a router, and each node is designed to sound its own alarm based on its sensors if a connection or sensor ever goes down.

We also picked up a few practical lessons along the way that are worth sharing: don't mess with validated code, double-check a module's address format before blaming the wiring, filter UWB distance readings since they tend to be noisy, and map out your pins carefully when swapping boards. Of course, the prototype still has its limitations. Most notably, the 1.5 m threshold is currently just a bench value the integrated field tests and UWB error measurements still need to be recorded, and the vibration motor stage isn't working quite yet. Once these final pieces are in place, this design provides a clear, affordable foundation for a proximity warning system that can be further developed for real-world mining conditions.

14. References

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15. Appendices

A: Circuit Diagram and Wiring Connections This section includes the complete circuit diagram showing how all the components are connected—including the ESP32, C4001 mm Wave radar, VL53L0X ToF sensor, ESP32-CAM, OLED display, buzzer, LEDs, and the power supply. It provides the finalized pin assignments and wiring details for each component.

B: Bill of Materials and Expenditure Details Here you will find a table listing every component used in the project, complete with names, quantities, specifications, unit prices, and total costs. This includes details of all components purchased and any replacements used during the testing phase.

C: ESP32 Program Code and Sensor Interfacing This section contains the final ESP32 source code used for reading the sensors, measuring distances, comparing thresholds, triggering alarms, updating the OLED display, and handling ESP-NOW communication. Brief explanations of the main functions and the sensor pin connections are included alongside the code.

D: UWB Module Specifications and Implementation Included here are the datasheet and technical specifications for the UWB module, along with its wiring diagram, interface details, and the implemented method for measuring the distance between the worker and the mobile equipment. The final calibration and testing results are attached.

E: Camera Module Configuration and Testing This section covers the ESP32-CAM configuration, camera settings, SD card setup, and the relevant program code. It includes the sample photos captured during testing and confirms that the image capture and storage functions operated correctly.

F: Photographs of the Prototype This section provides photographs of the assembled prototype, individual components, wiring connections, sensor placement, and the power supply arrangement. Each photo includes a caption describing what it shows and the stage of development it represents.

G: Sensor Testing and Alarm Response Here, the final test procedures and observations for the C4001 mm Wave radar and VL53L0X ToF sensor are documented. It records the measured detection distances, the configured alert threshold, how the buzzer and LEDs responded, and any limitations noticed during testing.

H: Wireless Communication and Power Budget This section includes the ESP-NOW communication setup, final test results, observed response delays, and any connection failures encountered. It also provides the estimated current consumption, power requirements, battery capacity, boost-converter efficiency, and calculated battery runtime, clearly distinguishing between estimated and experimentally measured results.

I: Component Datasheets and Technical Documents This section includes the relevant datasheets and manufacturer documentation for the ESP32, C4001 mm Wave radar, VL53L0X ToF sensor, ESP32-CAM, OLED display, UWB module, battery, TP4056 charging module, and boost converter. Only the documents for the components used in the prototype are included.

Annexures

Annexure A: Pin assignments

Machine node (ESP32)

Part	Pin(s)
C4001 radar TX / RX	GPIO16 (RX2) / GPIO17 (TX2)
VL53L0X SDA / SCL / XSHUT	GPIO21 / GPIO22 / GPIO23
OLED CLK / MOSI / CS / DC / RES	GPIO18 / GPIO19 / GPIO27 / GPIO26 / GPIO25
Buzzer / LED	GPIO14 / GPIO33

Worker node (ESP32-C3 Super Mini)

Part	Pin(s)
UWB module TXD to ESP32 RX / UWB module RXD from ESP32 TX	GPIO20 / GPIO21
UWB NRST	GPIO7
OLED SCK / SDA / RES / DC / CS (software SPI)	GPIO2 / GPIO4 / GPIO0 / GPIO1 / GPIO5
Buzzer / LED / vibration motor module	GPIO6 / GPIO3 / GPIO10

Camera node (ESP32-CAM)

Camera, SD slot and red LED (GPIO33, active low) are on the board; only 5 V, GND and the supply capacitor are wired externally.

Annexure B: ESP-NOW packet formats

Machine to camera and worker (10 bytes, magic 0xA5)

Field	Size (bytes)	Meaning
magic	1	0xA5
type	1	1 = sensor node, 2 = camera node
alert	1	1 when the machine's own alert is active
flags	1	Sensor: bit 0 human (radar), bit 1 obstacle (laser). Camera: bit 0 camera OK, bit 1 SD OK
value	2	Sensor: laser distance in mm (65535 = none). Camera: photos saved
seq	4	Message counter

Worker to machine (7 bytes, magic 0xA7)

Field	Size (bytes)	Meaning
magic	1	0xA7
type	1	2
seq	1	Message counter
uwbOk	1	1 when a fresh UWB distance exists
distCm	2	UWB distance in cm (65535 = none)
hazard	1	1 when the worker is inside the UWB alarm distance

Annexure C: Component list

Component	Qty	Used in
ESP32 DevKit / WROVER board	1	Machine node
ESP32-C3 Super Mini	1	Worker node
ESP32-CAM (AI-Thinker) with microSD card	1	Camera node
DFRobot C4001 mmWave radar	1	Machine node
VL53L0X laser distance sensor	1	Machine node
Reyax RYUW122 Lite UWB module	2	Worker node (anchor) and machine side (tag)
1.3 inch SPI OLED (SH1106)	2	Machine and worker nodes
Active buzzer module	2	Machine and worker nodes
Red LED with resistor	2	Machine and worker nodes
Vibration motor module	1	Worker node
18650 Li-ion 2s cell, MRB045 module, 7.4 V to 5 V buck converter	1 Set	Power
LiPo Battery, TP4056 Module, 3.7V to 5V boost converter	1 Set	Power
USB-to-serial adapter (FTDI)	1	Programming the ESP32-CAM
USB-toTTL Converter	1	Set the UWB Module as Tag and Anchor

Annexure D: Key configuration settings

Setting	Value
Radar mode and window	Speed mode, 30 to 300 cm, threshold 10, fretting detection on
Radar alert distance	150 cm or less
Laser alert distance	Below 1500 mm, read every 100 ms
Alarm hold time (machine)	2000 ms
UWB alarm on / off	150 cm / 170 cm, 2 consecutive readings
UWB distance filter	Median of last 5 readings
UWB lost after	5 failed polls
Machine status message	Every 100 ms, or immediately on a change
Worker status message	Every 200 ms
Link lost (machine and worker)	2000 ms without a message
Camera heartbeat / link timeout / photo cooldown	1000 ms / 1000 ms / 3000 ms
Wi-Fi channel	1 on all nodes
UWB network ID / addresses	REYAX123 / ANCHOR01 and TAG00001

Annexure E: UWB module commands used

Purpose	Command
Set role (1 = anchor, 0 = tag)	AT+MODE=1 or AT+MODE=0
Set network ID (8 characters)	AT+NETWORKID=REYAX123
Set module address (8 characters)	AT+ADDRESS=ANCHOR01 or AT+ADDRESS=TAG00001
Read settings back	AT+MODE?, AT+NETWORKID?, AT+ADDRESS?, AT+CPIN?
Measure distance to a tag	AT+ANCHOR_SEND=TAG00001,4,PING
Reply from the module	+ANCHOR_RCV=TAG00001,length,payload,distance cm